



NORTH SYDNEY BOYS HIGH SCHOOL

2012 YEAR 11 YEARLY EXAMINATION

Mathematics Extension 1

General Instructions

- Working time – 2.5 hours
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Attempt all questions
- Marks may be deducted for incomplete or illegible working

Class Teacher:

(Please tick or highlight)

- ☐ Mr Rezcallah/ Ms Juhn
- ☐ Mr Fletcher
- ☐ Ms Ziazaris
- ☐ Mr Lucas
- ☐ Mr Lam
- ☐ Mr Berry

Student Name:

(To be used by the exam markers only.)

Question No	1-10	11	12	13	14	15	16	17	18	Total	Total
Mark	$\frac{\quad}{10}$	$\frac{\quad}{14}$	$\frac{\quad}{11}$	$\frac{\quad}{11}$	$\frac{\quad}{10}$	$\frac{\quad}{14}$	$\frac{\quad}{12}$	$\frac{\quad}{13}$	$\frac{\quad}{15}$	$\frac{\quad}{110}$	$\frac{\quad}{100}$

Question 1

Evaluate $\frac{(7.12)^3 - \sqrt{8}}{\sqrt[3]{2}}$ to 3 significant figures:

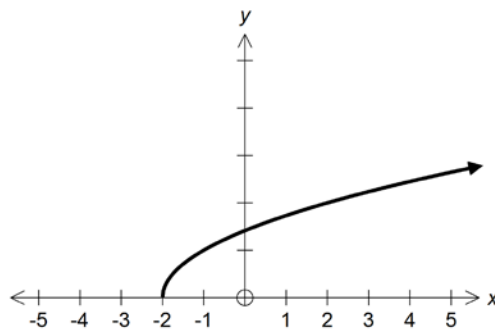
- a) 1.11 b) 84.4 c) 253 d) 284

Question 2

Which of the following is a function?

- a) $x = y^2$ b) $x = \frac{1}{y}$ c) $x^2 + y^2 = 1$ d) $\{1,2\}, \{2,5\}, \{1,-4\}$

Question 3



The function above can be best described as:

- a) $y = \sqrt{x-2}$ b) $y = \sqrt{x+2}$ c) $y = \sqrt{x^2-2}$ d) $y = \sqrt{x^2+2}$

Question 4

If $\sin \theta = \frac{2}{5}$ and $\cos \theta < 0$, which of the following is the value of $\tan \theta$?

- a) $-\frac{\sqrt{21}}{2}$ b) $-\frac{2}{\sqrt{21}}$ c) $\frac{2}{\sqrt{21}}$ d) $\frac{\sqrt{21}}{2}$

Question 5

The perpendicular distance of (2, 3) from $2x - y + 3 = 0$ is given by:

- a) $\frac{\sqrt{5}}{4}$ b) $\frac{\sqrt{13}}{4}$ c) $\frac{4}{\sqrt{13}}$ d) $\frac{4}{\sqrt{5}}$

Question 6

Which of the following is not a test for congruent triangles?

- a) SAS b) SSA c) AAS d) SSS

Question 7

Solve for x : $7^x = 200$.

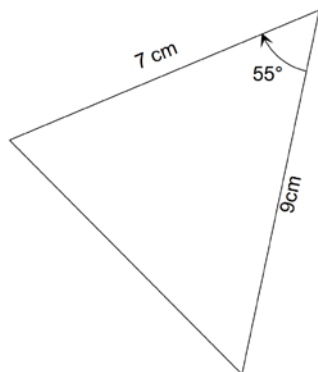
- a) 0.328 b) 0.367 c) 2.72 d) 237

Question 8

The derivative of $\frac{2}{x^3}$ is:

- a) $\frac{2x^2}{3}$ b) $-\frac{6}{x^4}$ c) $\frac{6}{x^2}$ d) $6x^4$

Question 9



The area of the above triangle is given by:

- a) $\frac{63 \sin 55^\circ}{2}$ b) $63 \sin 55^\circ$ c) $\frac{63 \cos 55^\circ}{2}$ d) $63 \cos 55^\circ$

Question 10

The sum of the first n terms of an arithmetic series with a first term of 10 and a common difference of -2 is given by:

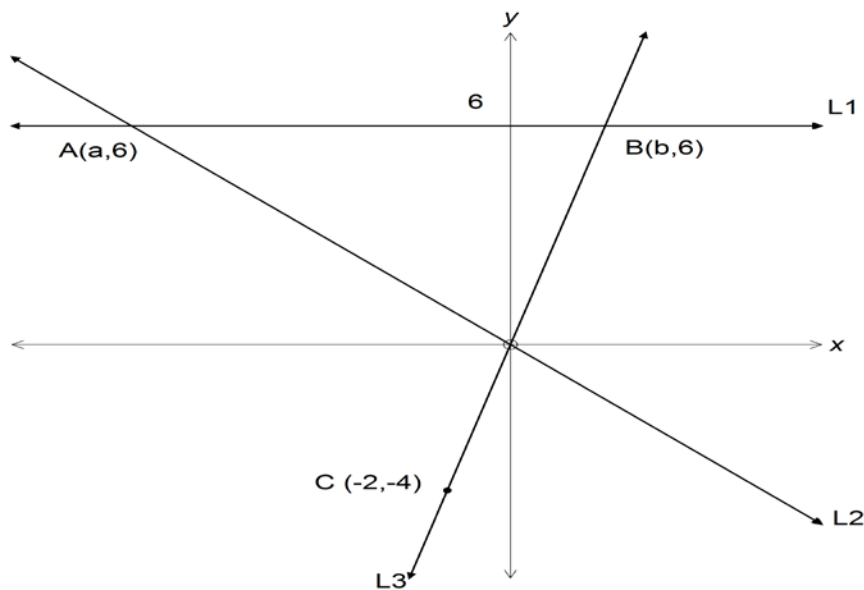
- a) $3 - 2n$ b) $7 - 2n$ c) $5n - 2n^2$ d) $11n - n^2$

Question 11 (14 Marks)

- (a) Express $\frac{5}{\sqrt{5}-\sqrt{3}}$ with a rational denominator. 2
- (b) Solve for x : $|3x + 2| = 5$ 2
- (c) Solve $\frac{4x-3}{x} \geq 5$ and sketch your solution on a number line. 3
- (d) Clearly sketch the region specified by: $(x-5)^2 + (y+4)^2 < 4$ 3
- (e) i) Sketch $f(x) = x^2 - 3x - 4$ showing all relevant features. 2
- ii) Hence, or otherwise, sketch $f(x) = \frac{1}{x^2 - 3x - 4}$ on a separate number plane. 2
(do not use calculus)

Question 12 (11 Marks)

- (a) In the diagram below, the line $L1$ is parallel with the x -axis and crosses the y -axis at 6. Lines $L2$ and $L3$ each pass through the origin, O , and intersect with $L1$ at the points $A(a,6)$ and $B(b,6)$ respectively. The point $C(-2,-4)$ lies on the line $L3$.



- i) Show that the equation of the line $L3$ is $2x - y = 0$ 2
- ii) Show that the x -ordinate of point B is 3. 1
- iii) Find the coordinates of point A such that $\angle AOB$ is a right angle 3
- iv) Write down the distance between points A and B . Hence, or otherwise calculate the area of $\triangle AOB$. 2
- (b) $P(0, 0)$ divides AB externally in the ratio $2 : 1$, where A is (x, y) and B is $(4, 9)$. Find the values of x and y . 3

Question 13 (11 Marks)

(a) Differentiate the following with respect to x and simplify your answer:

i) $y = x^2\sqrt{x^3}$ 2

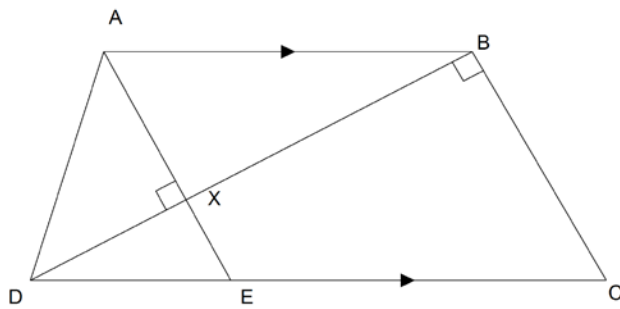
ii) $y = x^2(x - 5)^3$ 3

iii) $y = \frac{x^3+2}{5-x}$ 3

(b) Find the equation of the tangent to the curve $y = \sqrt{x+2}$ at the point where $x = 5$. 3

Question 14 (10 Marks)

(a) $ABCD$ is a trapezium where $AB \parallel DC$. Also BD is perpendicular to BC and AE is perpendicular to BD at X .



i) Copy the diagram into your writing booklet and find AX given that $AD = 41$ cm and $DX = 9$ cm. 1

ii) What type of quadrilateral is $ABCE$. Give reasons. 2

iii) Show that $\triangle DXE \parallel \triangle DBC$ 2

iv) Hence show that $BX \cdot XE = 360$. 3

(b) Draw a neat sketch of $y = \log_2(x - 1)$ clearly showing all relevant features. 2

Question 15 (14 Marks)

(a) Consider the curve $f(x) = -\frac{1}{3}x^3 - x^2 + 3x + 1$

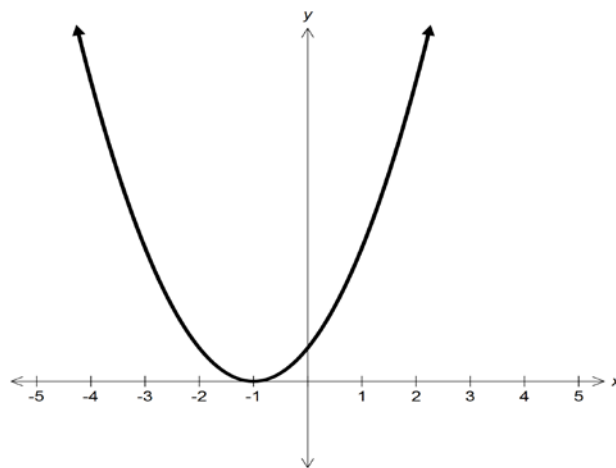
(i) Find the coordinates of any stationary points and determine their nature. 4

(ii) Find any point(s) of inflexion. 2

(iii) Sketch the curve in the domain $-6 \leq x \leq 3$. 2
(You do not need to find the x intercepts)

(iv) What is the maximum value of $f(x)$ in the given domain? 1

(b) Consider the function $y = f(x)$, whose gradient function is shown as a parabola in the sketch below.



(i) Comment on the sign of the gradient function for all $x \neq -1$ 1

(ii) What can you conclude about $y = f(x)$ when $x = -1$ 1

(iii) Sketch a graph for $y = f''(x)$ 1

(iv) Sketch a possible graph for $y = f(x)$ 2

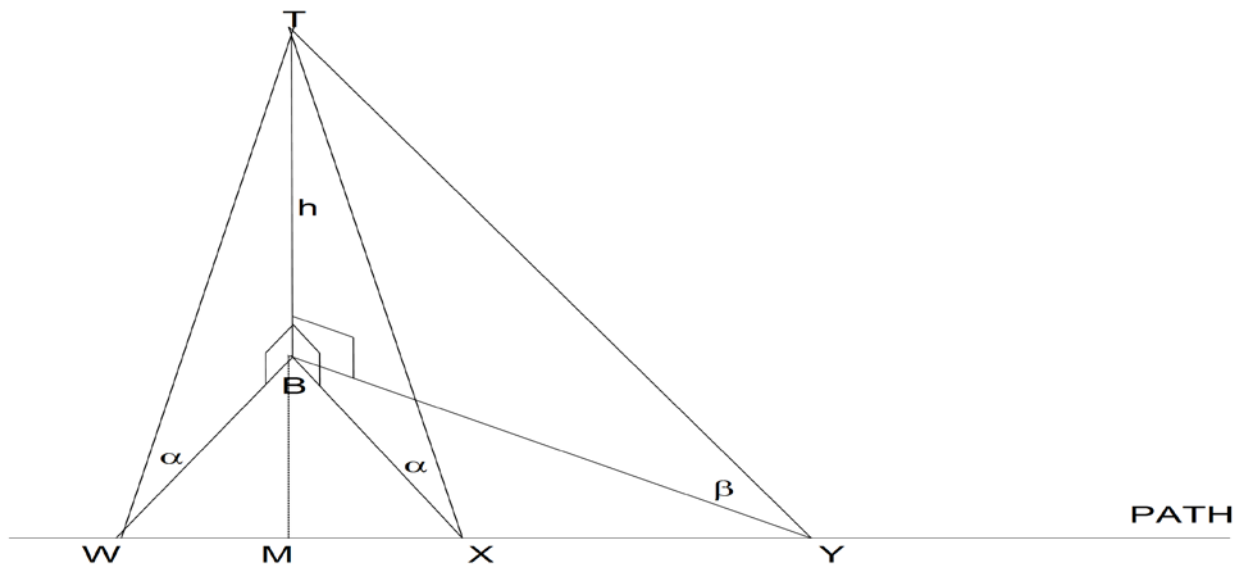
Question 16 (12 Marks)

(a) In $\triangle ABC$, $AC = 8$ cm, $BC = 10$ cm and $\angle ABC = 50^\circ$. Find the $\angle BAC$ to the nearest minute. 3

(b) Prove that $(\cot \theta + \operatorname{cosec} \theta)^2 = \frac{1 + \cos \theta}{1 - \cos \theta}$ 3

Question 16 Continued

- (c) Three points W , X and Y lie on a straight level path, where $WX = XY = 40$ m. The base, B , of a flagpole, TB , is level with the path. M is the midpoint of WX . The angles of elevation to the top of the flagpole from the points W , X and Y are α , α and β respectively. The height of the flagpole is h .



- i) Prove that $\triangle BWX$ is isosceles. 2
- ii) Find the length of BM in terms of h and α . 1
- iii) Hence, or otherwise, show that the height of the flagpole is given by:

$$h = \frac{40\sqrt{2}}{\sqrt{\cot^2 \beta - \cot^2 \alpha}} \quad 3$$

Question 17 (13 Marks)

- (a) The first and last terms of an arithmetic series are 10 and 66, respectively, and the sum of the series is 570. Find:

- i) The number of terms in the series. 2
- ii) The common difference 2

- (b) The general term of a geometric series is defined by $T_n = (x - 2)^n$. Find:

- i) The value(s) of x for which the series has a limiting sum: 1
- ii) This limit in terms of x . 2

- (c) For the series $0 + \log_{10} x^2 + \log_{10} x^4 + \dots$

- i) Prove that this series is arithmetic and hence find the common difference. 2
- ii) Hence, or otherwise, find the sum of the first 10 terms. 1

- (d) Find the value of x if $\sum_{n=0}^{\infty} \frac{9}{x^{n+1}} = 18$ 3

Question 18 (15 Marks)

(a) A particle moves in such a way that its distance, x metres, from the origin after t seconds is given by $x = 3 + 18t - t^3$ for $t \geq 0$.

i) Find an equation for its velocity after t seconds. 1

ii) At what time does the particle stop? 1

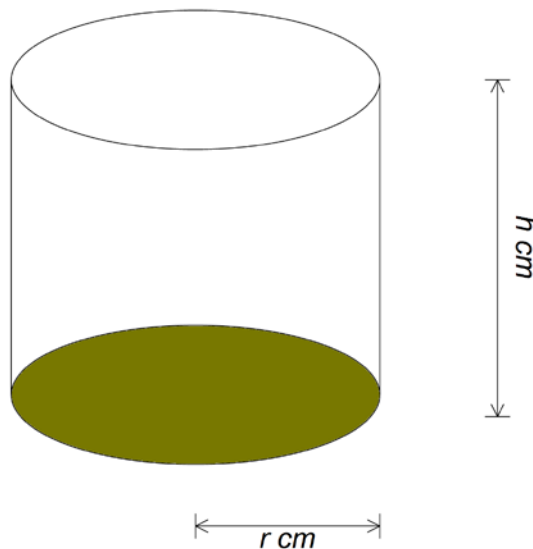
iii) Where is the particle initially? 1

iv) Find the velocity after 1 second. 1

v) How far has the particle travelled in the first second? 1

vi) What is the particle's acceleration after 2 seconds? 2

(b) A cylindrical can with an open top has a base radius of r cm and a height of h cm. It is to be made using 75π cm² of tin.



(i) Show that $h = \frac{75 - r^2}{2r}$ 2

(ii) Find an expression for the volume of the can, V , in terms of r . 1

(iii) Show that the maximum volume of this can is 125π cm³. 4

(iv) Find the height of the can from (iii). 1

End of examination

1 mark each

1: D

2: B

3: B

4: B

5: D

6: B

7: C

8: B

9: A

10: D. #

(11)

(a) i) $\frac{5}{\sqrt{5}-\sqrt{3}} \times \frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}+\sqrt{3}}$ ✓

$$= \frac{5(\sqrt{5}+\sqrt{3})}{2}$$
 ✓

(b) $|3x+2| = 5$

$$\therefore 3x+2 = 5$$

$$3x = 3$$

$$x = 1$$
 ✓

or

$$3x+2 = -5$$

$$3x = -7$$

$$x = -\frac{7}{3}$$
 ✓

method 1

(c) $\frac{4x-3}{x} \geq 5$

$$x \neq 0$$

$$\frac{4x-3}{x} = 5$$

$$4x-3 = 5x$$

$$\therefore x = -3$$

or

method 2

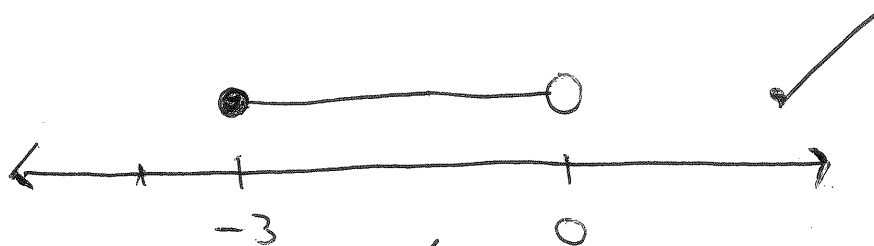
$$\frac{4x-3}{x} \times x^2 \geq 5 \times x^2$$
 ✓

$$4x^2 - 3x \geq 5x^2$$

$$\therefore x^2 + 3x \leq 0$$

$$x(x+3) \leq 0$$

$$\therefore -3 \leq x < 0, \text{ as } x \neq 0$$
 ✓



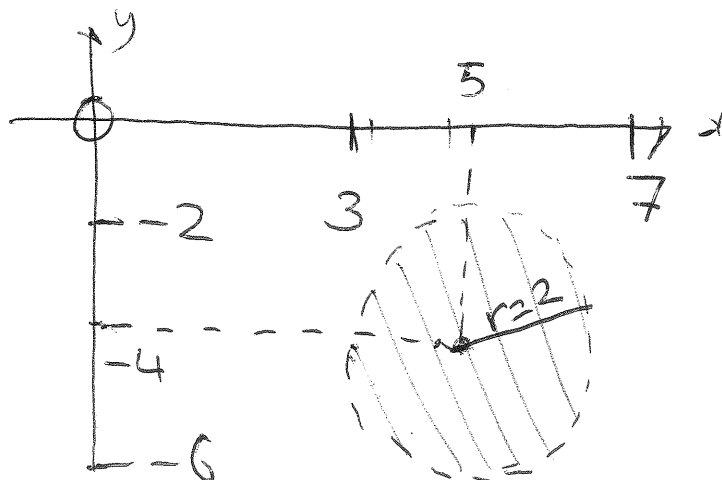
x

x

← testing a point in each region

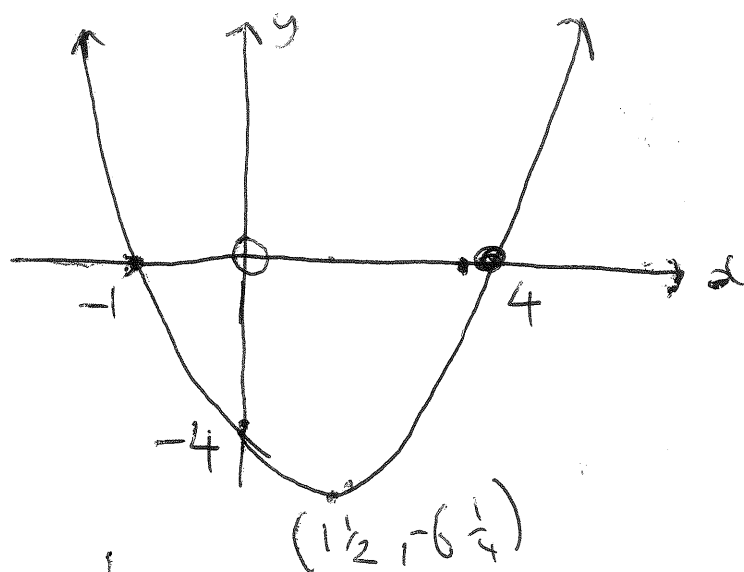
$$\therefore -3 \leq x < 0$$

d)



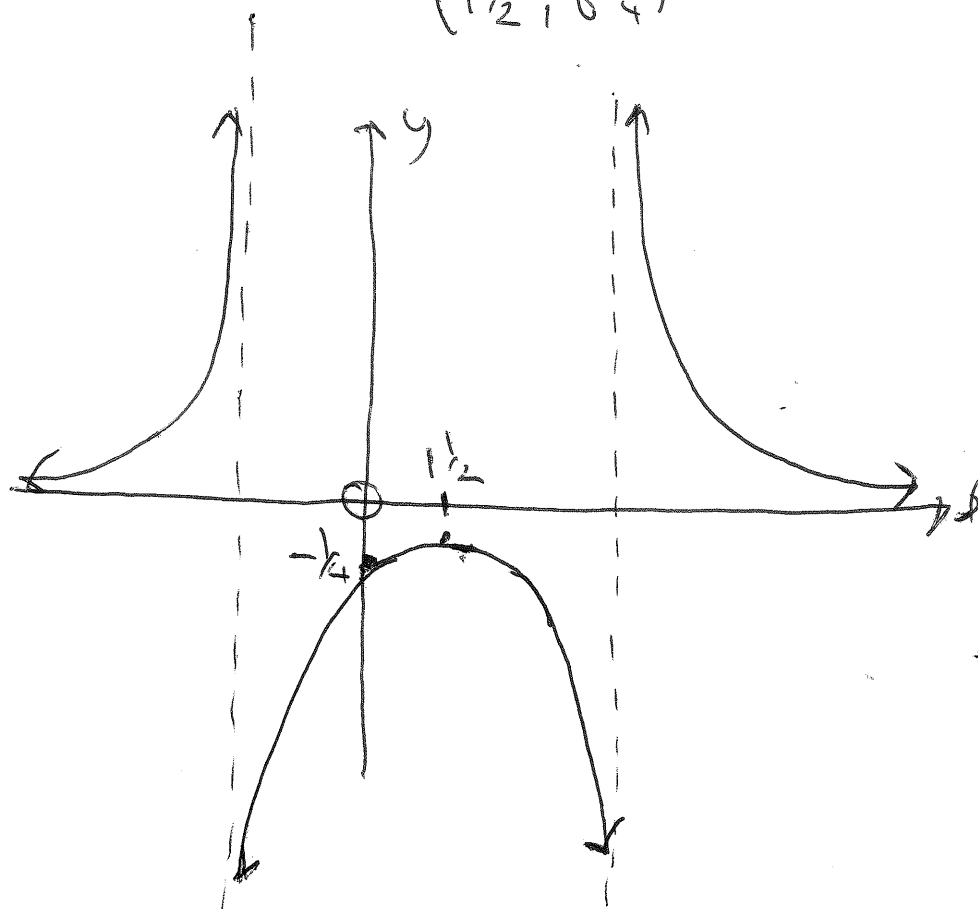
- ✓ dotted line
- ✓ shade region
- ✓ correct circle

e) i) $y = (x-4)(x+1)$



- ✓ intercepts
- ✓ vertex
(0 if wrong shape)

ii)



- ✓ shape
- ✓ asymptotes/
point

(12)

(a) i) $\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$

$\therefore \frac{y - 0}{x - 0} = \frac{-4 - 0}{-2 - 0}$

$\therefore -2y = -4x$

$\therefore 4x - 2y = 0$

$2x - y = 0$, as req'd ✓

ii) when $y = 6$
 $2x - 6 = 0$ (eqn of L3) ✓
 $2x = 6$
 $x = 3$

iii) m of L3 = 2. ✓

$\therefore m_{L3} \times m_{L2} = -1$

$\therefore m_{L2} = -1/2$

$\therefore y = -1/2 x$ (L2 goes through origin) ✓
when $y = 6$

$6 = -1/2 \times x$
 $\therefore x = -12$ ✓

iv) $AB = 15$ units. ✓

$AOB = \frac{1}{2} \times 15 \times 6$
 $= 45 \text{ units}^2$ ✓

$$\begin{aligned}
 \textcircled{b} \quad x: \quad 0 &= \frac{m x_2 + n x_1}{m + 2} \\
 &= \frac{(2)(4) + (-1)(x)}{2 + (-1)} \\
 &= 8 - x
 \end{aligned}$$

$$\therefore x = 8$$

$$\begin{aligned}
 y: \quad 0 &= \frac{m y_2 + n y_1}{m + n} \\
 &= \frac{(2)(9) + (-1)(y)}{2 + (-1)} \\
 &= 18 - y
 \end{aligned}$$

$$\therefore y = 18$$

$$\therefore A(x, y) \text{ is } (8, 18)$$

(full marks if they do a diagram)
(and solve by inspection)

(b)

$$y = (x+2)^{\frac{1}{2}}$$

$$y' = \frac{1}{2} (x+2)^{-\frac{1}{2}}$$

- 1

$$m = \frac{1}{2\sqrt{7}}$$

- 1

$$x=5 \quad y=\sqrt{7}$$

$$y-y_1 = m(x-x_1)$$

$$y-\sqrt{7} = \frac{1}{2\sqrt{7}}(x-5)$$

$$2\sqrt{7}(y-\sqrt{7}) = x-5$$

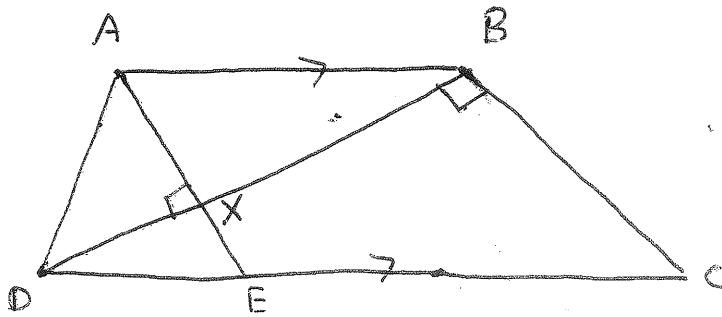
$$2\sqrt{7}y - 14 = x - 5$$

$$x - 2\sqrt{7}y + 9 = 0$$

- 1

14

Q. 1)



$$(AX)^2 = (AD)^2 - (DX)^2 \quad (\text{Pythagoras})$$

$$\therefore AX = \sqrt{41^2 - 9^2}$$

$$= 40 \text{ cm}$$

ii) $AB \parallel EC$ (given) ✓
 $AE \parallel BC$ (alt $\angle$ s both 90°) ✓
 $\therefore$ parallelogram (opposite sides parallel) ✓

iii) In ΔDXE & ΔDBC
 $\angle DXE = \angle DBC$ (both 90°) ✓
 $\angle XDE = \angle BDC$ (common)
 $\therefore \Delta DXE \sim \Delta DBC$ (equiangular) ✓

iv) $\frac{XE}{BC} = \frac{DX}{DB}$ (matching sides of similar Δ s in ratio) ✓
 (for $BC=AE$)

$$\therefore XE, DB = BC, DX$$

$$\therefore XE(BX + 9) = 9(XE + 40) \quad (\text{since } BC = AE, \text{ opp sides of } \text{pyram equal})$$

$$\therefore BX \cdot XE + 9(XE) = 9(XE) + 360$$

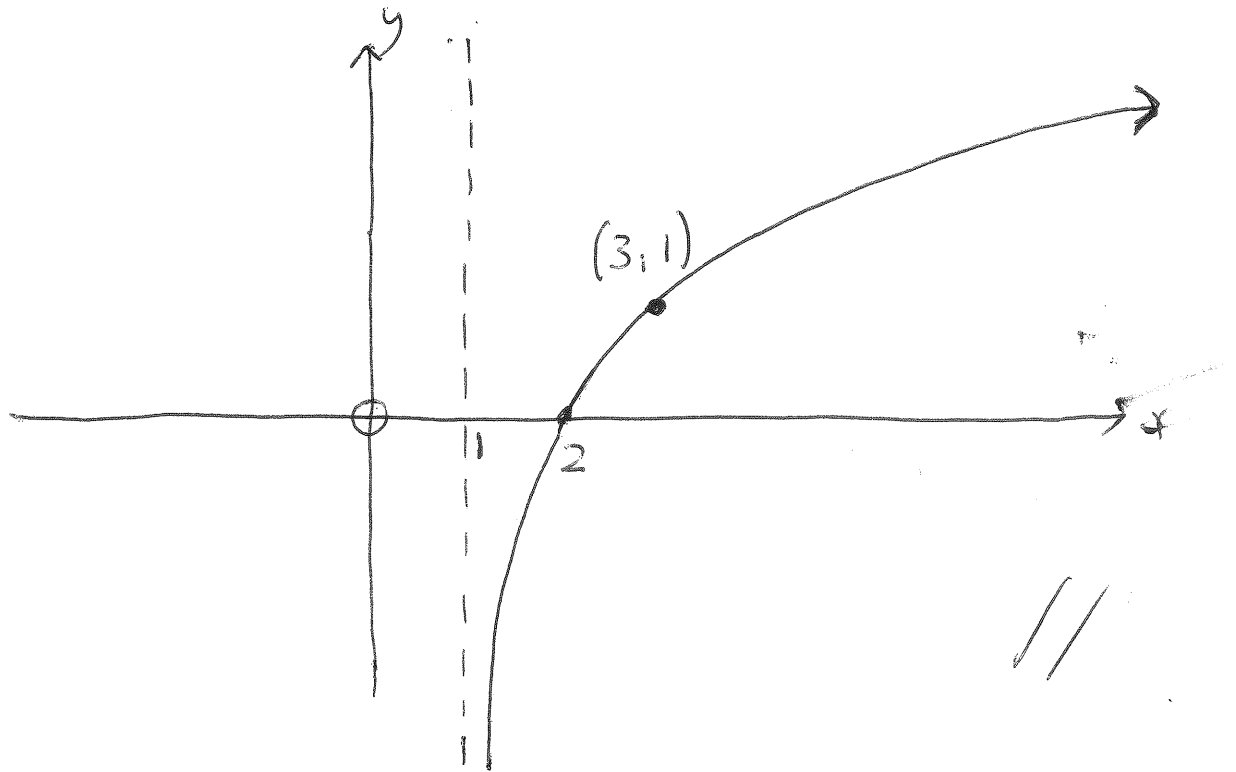
$$\therefore BX, XE = 360, \text{ as req'd.} \quad \checkmark$$

alternately = Prove $\Delta AXB \sim \Delta EXD$

$$\text{Then } \frac{EX}{AX} = \frac{DX}{BX} \Rightarrow \frac{XE}{40} = \frac{9}{BX}$$

$$BX \cdot XE = 360$$

⑥



* -1 for each mistake

* full marks even if $(3, 1)$ not labelled,

15

9) i) $f(x) = -\frac{1}{3}x^3 - x^2 + 3x + 1$

$f'(x) = -x^2 - 2x + 3$ ✓

$= -(x^2 + 2x - 3)$

$= -(x+3)(x-1)$

∴ s.p.s at -3 and 1 ✓

x	-4	-3	0	1	2
$f'(x)$	-	0	+	0	-
	↖	-	↗	-	↘

∴ minima at $(-3, -8)$ ✓

maxima at $(1, 2\frac{2}{3})$ ✓

ii) $f''(x) = -2x - 2$

$f''(x) = 0$

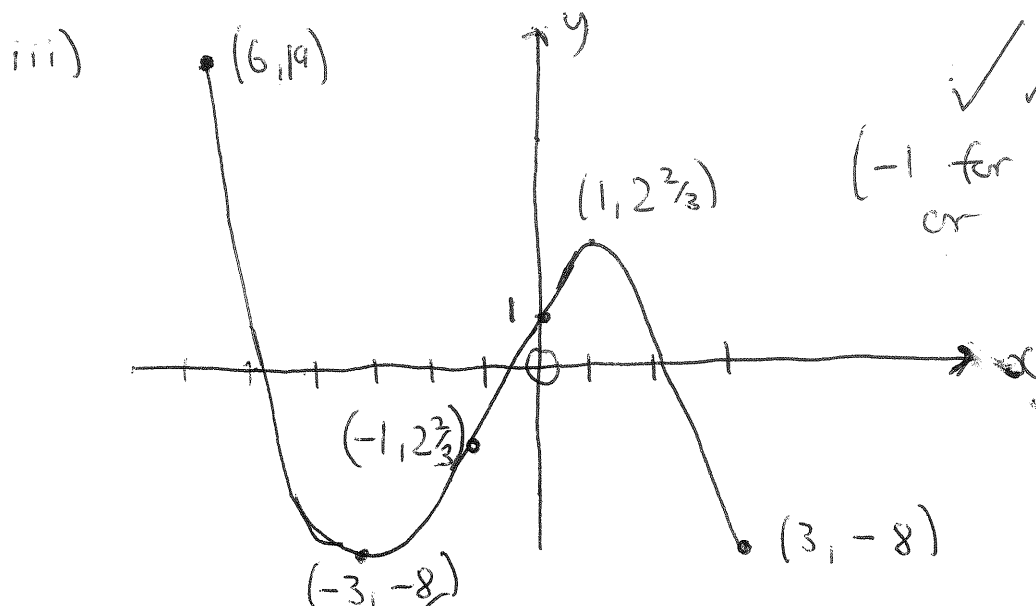
$2x = -2$ ✓

$x = -1$ ✓

x	-2	-1	0
$f''(x)$	+	0	-
	∪	.	∩

Concavity change means P.O.I

∴ P.O.I at $(-1, -2\frac{2}{3})$



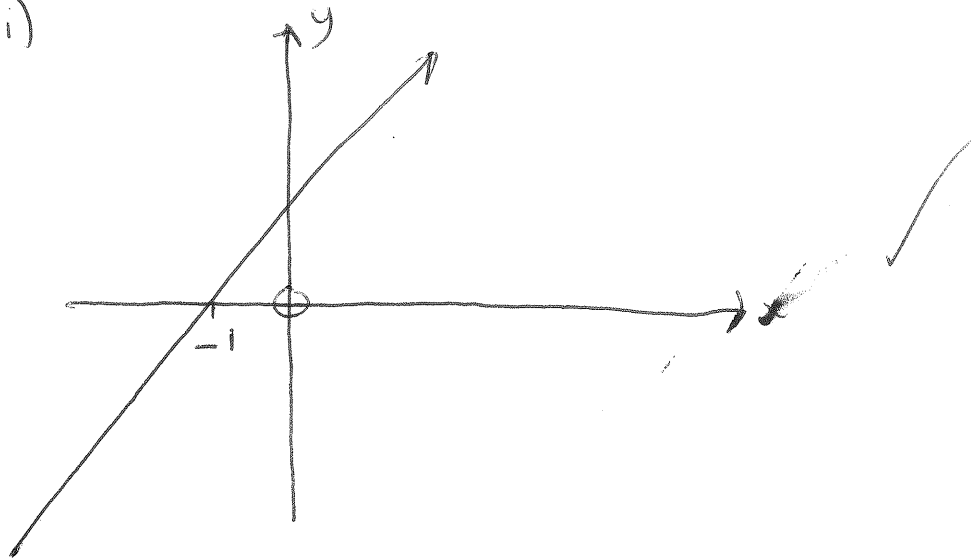
✓✓
(-1 for each mistake or omission)

iv) $y = 19$ ✓

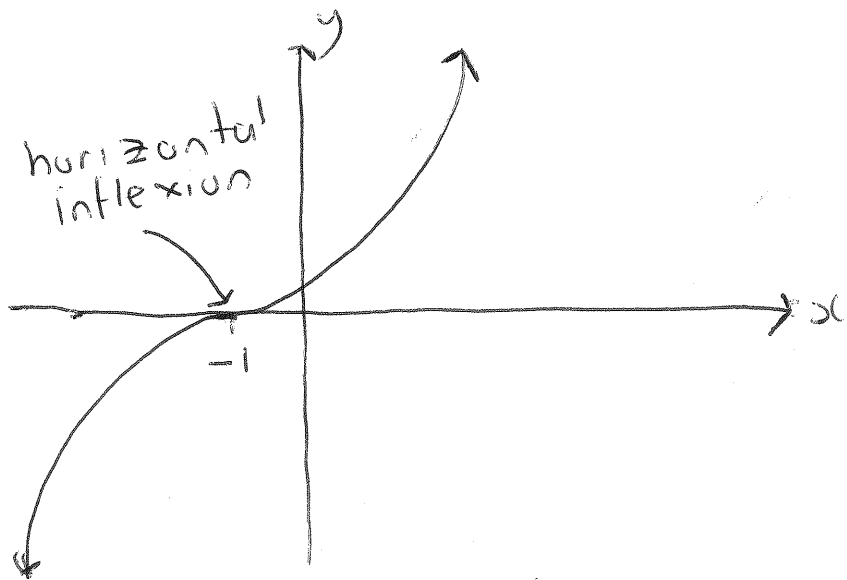
⑥ i) the gradient is positive ✓

ii) there is a point of horizontal inflexion. ✓ must say 'horizontal' and 'inflexion'

iii)



iv)

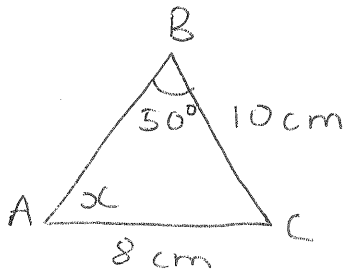


✓ S.P.

✓ inflexion.

16.

(a)



$$\frac{\sin x}{10} = \frac{\sin 50^\circ}{8} \quad \checkmark$$

$$\therefore x = \sin^{-1} \left(\frac{10 \sin 50^\circ}{8} \right)$$

$$\therefore x = 73^\circ 15' \quad \checkmark \quad \text{or } 106^\circ 45' \quad \checkmark$$

(b)

$$\text{LHS} = (\cot \theta + \operatorname{cosec} \theta)^2$$

$$= \left(\frac{\cos \theta}{\sin \theta} + \frac{1}{\sin \theta} \right)^2 \quad \checkmark$$

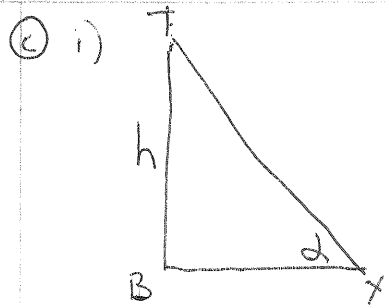
$$= \frac{(\cos \theta + 1)^2}{\sin^2 \theta}$$

$$= \frac{(\cos \theta + 1)^2}{1 - \cos^2 \theta} \quad \checkmark$$

$$= \frac{(\cancel{\cos \theta} + 1)(\cos \theta + 1)}{(1 + \cancel{\cos \theta})(1 - \cos \theta)}$$

$$= \frac{1 + \cos \theta}{1 - \cos \theta} \quad \checkmark$$

$$= \text{RHS, as required}$$

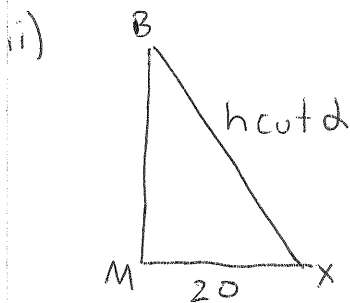


$$\cot \alpha = \frac{BX}{h}$$

$$\therefore BX = h \cot \alpha \quad \checkmark$$

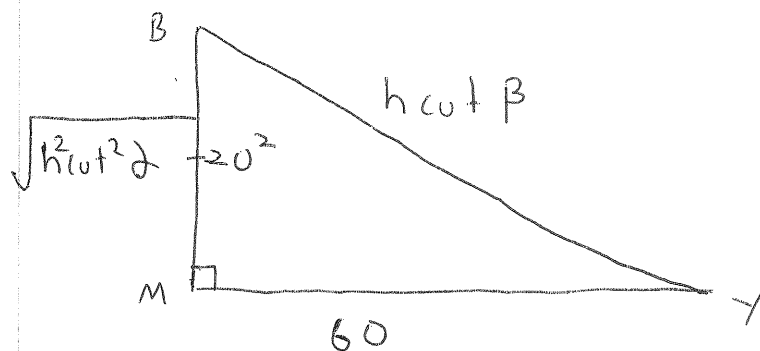
$$\text{similarly in } \triangle BTW, BW = h \cot \alpha \quad \checkmark$$

$$\therefore \triangle BWX \text{ is isosceles as } BW = BX$$



$$\therefore BM = \sqrt{h^2 \cot^2 \alpha - 20^2} \quad \checkmark$$

$$\text{iii) } BY = h \cot \beta \quad \checkmark$$



$$60^2 + h^2 \cot^2 \alpha - 20^2 = h^2 \cot^2 \beta \quad \checkmark$$

$$\therefore h^2 \cot^2 \alpha + 3200 = h^2 \cot^2 \beta$$

$$h^2 \cot^2 \beta - h^2 \cot^2 \alpha = 3200$$

$$h^2 (\cot^2 \beta - \cot^2 \alpha) = 3200$$

$$\therefore h^2 = \frac{3200}{\cot^2 \beta - \cot^2 \alpha}$$

$$\therefore h = \sqrt{\frac{3200}{\cot^2 \beta - \cot^2 \alpha}}$$

$$\therefore h = \frac{40\sqrt{2}}{\sqrt{\cot^2 \beta - \cot^2 \alpha}}$$



(17)

a) i) $S_n = \frac{n}{2} (a + l)$ ✓

$\therefore 570 = \frac{n}{2} (10 + 66)$ ✓

$\therefore 76n = 1140$

$\therefore n = 15$ ✓

ii) $T_n = a + (n-1)d$ ✓

$66 = 10 + (15-1)d$

$\therefore 14d = 56$

$\therefore d = 4$ ✓

b) i) $r = \frac{(x-2)^{n+1}}{(x-2)^n}$

$r = x-2$

$\therefore -1 < x-2 < 1$

$\therefore 1 < x < 3$ ✓

ii) $S_\infty = \frac{a}{1-r}$ ✓

$= \frac{x-2}{1-(x-2)}$

$= \frac{x-2}{3-x}$ ✓

$$\begin{aligned} \textcircled{c} \text{ i) } T_3 - T_2 &= \log_{10} x^4 - \log_{10} x^2 \\ &= 2 \log_{10} x^2 - \log_{10} x^2 \\ &= \log_{10} x^2 \end{aligned}$$

✓ proof

$$\begin{aligned} T_2 - T_1 &= \log_{10} x^2 - 0 \\ &= \log_{10} x^2 \end{aligned}$$

$$\checkmark d = \log_{10} x^2$$

$$\therefore T_3 - T_2 = T_2 - T_1$$

$\therefore$ this is an AP where $d = \log_{10} x^2$

$$\begin{aligned} \text{ii) } S_n &= \frac{n}{2} (2a + (n-1)d) \\ &= \frac{10}{2} (2 \times 0 + (10-1) \log_{10} x^2) \\ &= 5 \times 9 \log_{10} x^2 \\ &= 45 \log_{10} x^2 \\ &= 90 \log_{10} x \text{ or } \log_{10} x^{90} \end{aligned}$$

✓

$\textcircled{d}$ this is a GP where:

$$n = \infty$$

$$a = \frac{9}{x}$$

$$r = \frac{1}{x}$$

$$S_n = 18$$

$$S_n = \frac{a}{1-r}$$

✓

$$\frac{\frac{9}{x}}{1 - \frac{1}{x}} = 18$$

✓

$$\therefore \frac{9}{x-1} = 18$$

$$\therefore 9 = 18x - 18$$

$$\therefore \cancel{18x} - 18x = 36 - 18$$

$$\therefore 18x = 36$$

$$\therefore x = \frac{3}{2}$$

✓

Suggested Solutions

Question 18 (Lam)

(a) $x = 3 + 18t - t^3$

i. (1 mark)

$$\frac{dx}{dt} = 18 - 3t^2$$

ii. (1 mark)

$$\frac{dx}{dt} = 18 - 3t^2 = 0$$

$$3(6 - t^2) = 0$$

$$t^2 = 6$$

$$\therefore t = \sqrt{6}$$

as $t > 0$.

iii. (1 mark)

$$x = 3$$

iv. (1 mark)

$$\left. \frac{dx}{dt} = 18 - 3t^2 \right|_{t=1} = 15 \text{ ms}^{-1}$$

v. (1 mark)

$$\begin{aligned} x &= 3 + 18t - t^3 \Big|_{t=1} \\ &= 3 + 18 - 1 = 20 \text{ m} \end{aligned}$$

Hence particle travels $20 - 3 = 17$ m in the first second (see part (iii))

vi. (2 marks)

✓ [1] for correct second derivative.
Fatal error if this is not done properly.

✓ [1] for proper substitution and correct answer.

$$\left. \frac{d^2x}{dt^2} = -6t \right|_{t=2} = 12 \text{ ms}^{-2}$$

(b) i. (2 marks)

✓ [1] for progress towards answer.

✓ [2] for properly set out answer.

$$A = \pi r^2 + 2\pi rh = 75\pi$$

$$\pi(r^2 + 2rh) = 75\pi$$

$$r^2 + 2rh = 75$$

$$2rh = 75 - r^2$$

$$\therefore h = \frac{75 - r^2}{2r}$$

ii. (1 mark)

$$\begin{aligned} V &= \pi r^2 h = \pi r^2 \left(\frac{75 - r^2}{2r} \right) \\ &= \frac{\pi}{2} r (75 - r^2) = \frac{\pi}{2} (75r - r^3) \end{aligned}$$

iii. (4 marks)

✓ [1] for proper differentiation.

✓ [1] for $r = 5$ through showing working.

✓ [1] for testing local maximum.

✓ [1] for fully showing $V = 125\pi$.

$$\frac{dV}{dr} = \frac{\pi}{2} (75 - 3r^2)$$

Stationary points when $\frac{dV}{dr} = 0$:

$$\frac{\pi}{2} (75 - 3r^2) = 0$$

$$\therefore 75 - 3r^2 = 0$$

$$3r^2 = 75$$

$$r^2 = 25$$

$$\therefore r = 5$$

as $r > 0$. Finding the second derivative,

$$\frac{d^2V}{dr^2} = \frac{\pi}{2} \times -6r = -3\pi r$$

When $r = 5$,

$$\frac{d^2V}{dr^2} < 0$$

 $\therefore r = 5$ is a local maximum. Hence the maximum volume is

$$V = \frac{\pi}{2} \times 5 (75 - 25) = 125\pi$$

iv. (1 mark)

$$h = \frac{75 - 5^2}{2 \times 5} = 5$$